



## Sheet 6

### Relativistic Dynamics

#### Relativistic Momentum and The Relativistic Form of Newton's Laws

1. Calculate the momentum of a proton and an electron moving with speed of  $0.90c$ . Convert the answer to  $\text{MeV}/c$ .
2. Consider the relativistic form of Newton's second law. Show that when  $F$  is parallel to  $v$ ,

$$F = \gamma^3 ma$$

where  $m$  is the mass of an object and  $a$  is its acceleration.

#### Relativistic Energy

3. Show that the energy–momentum relationship given by  $E^2 = p^2c^2 + (mc^2)^2$  follows from the expressions  $E = \gamma mc^2$  and  $p = \gamma mu$ .
4. An electron has a speed  $u = 0.850c$ . Find its total energy and kinetic energy in electron volts.
5. The total energy of a proton is three times its rest energy. (a) Find the proton's rest energy in electron volts. (b) With what speed is the proton moving? (c) Determine the kinetic energy of the proton in electron volts. (d) What is the proton's momentum?
6. A proton in a high-energy accelerator is given a kinetic energy of  $50 \text{ GeV}$ . Determine the (a) momentum and (b) speed of the proton.
7. An electron has a speed of  $0.75c$ . Find the speed of a proton that has (a) the same kinetic energy as the electron and (b) the same momentum as the electron.

## Conservation of Relativistic Momentum and Energy

8. A radium isotope decays to a radon isotope,  $^{222}\text{Rn}$ , by emitting an  $\alpha$  particle (a helium nucleus) according to the decay scheme  $^{226}\text{Ra} \rightarrow ^{222}\text{Rn} + ^4\text{He}$ . The masses of the atoms are 226.0254 (Ra), 222.0175 (Rn), and 4.0026 (He). How much energy is released as the result of this decay?
9. Calculate the binding energy in MeV per nucleon in the isotope  $^{12}\text{C}_6$ . Note that the mass of this isotope is exactly 12 u, and the masses of the proton and neutron are 1.007276 u and 1.008665 u, respectively.
10. An electron having kinetic energy  $K = 1.000$  MeV makes a head-on collision with a positron at rest. (A positron is an antimatter particle that has the same mass as the electron but opposite charge.) In the collision the two particles annihilate each other and are replaced by two  $\gamma$  rays of equal energy, each traveling at equal angles  $\theta$  with the electron's direction of motion. Find the energy  $E$ , momentum  $p$ , and angle of emission  $\theta$  of the  $\gamma$  rays.